** Types of Graphical Scale:-

→ plain Scale
→ Diagonal Scale
→ Vernier Scale.

* plain scale:-

→ A plain scale consists of a line divided into suitable number of equal units. The first unit is subdivided into smaller parts.

→ The zero should be placed at the end of I^{st} main unit.

→ From the zero mark, the units should be numbered to the right and the sub-divisions to the left.

→ The R.F should be mentioned below the scale.

Ex:- Construct a scale of 1:4, to show centimeters and long enough to measure upto 5 decimeters.

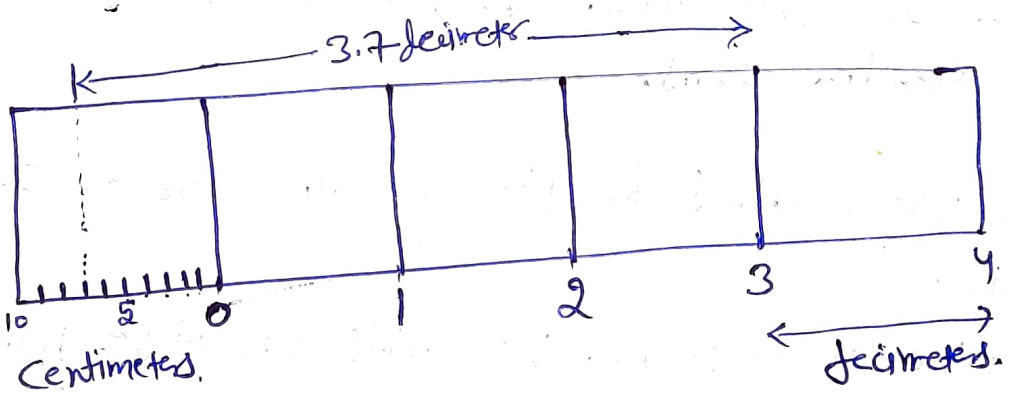
- R.F = 1/4
- Length of the scale = R.F × max^m length
  = 1/4 × 5 dm = 12.5 cm.
- Draw a line of 12.5 cm long and divide it into 5 equal division, each represents 1 dm.

~~Then divide this line into~~

- Mark 0 at the end of the first division and 1, 2, 3 and 4 at the end of each subsequent division to its right.

→ Divide the first division to 10 equal sub-division, each representing 1 cm.

→ Mark cm to the left of 0 as shown.



* Diagonal Scale:-

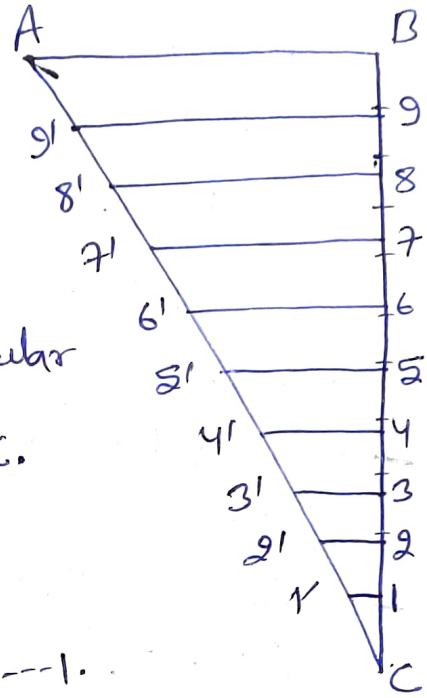
→ Through Diagonal Scale, measurements can be upto second decimal places (eg- 4.35).

→ Diagonal Scales are used to measure distances in a unit and its immediate two subdivisions.

eg:- dm, cm & mm or yard, foot, inch

→ Diagonal Scale can measure more accurately than the plane scale.

⇒ Diagonal Scale Concept:-



→ At end B of line AB, draw a perpendicular.

→ Step-off ten equal division of any length along the perpendicular starting from B and ending at C.

→ Join A with C.

→ Number the division point 9, 8, 7, ---1.

→ Through the point 1, 2, 3, etc. draw lines parallel to AB and cutting AC at 1', 2', 3', etc.

→ Since the triangle are similar;

1'1 = 0.1 AB , 2'2 = 0.2 AB -----, 9'9 = 0.9 AB.

→ gives division of a given short lin AB in multiples of 1/10 its length, eg. 0.1 AB, 0.2 AB, 0.3 AB, etc.

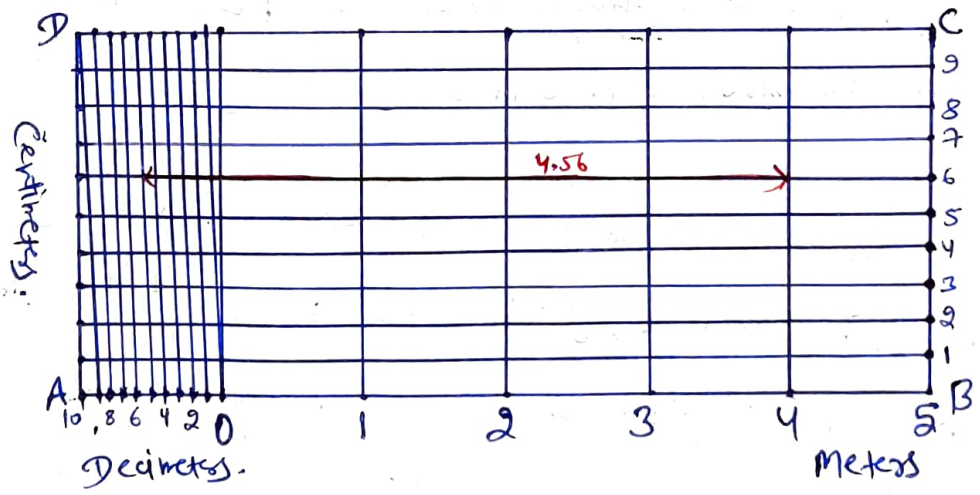
e.g:- Construct a Diagonal Scale of RF = 3:200 Showing meters, decimeters and centimeters. The Scale should measure up to 6 m. Show a distance of 4.56 m.

—x—x—x—x—

Now - length of Scale = $\frac{3}{200} \times 6m = 9$ cm.

- Draw a line AB = 9 cm and devide it in to 6 equal parts.

- Divide the first part AO into 10 equal parts.
- At A draw a perpendicular and step off along it 10 equal divisions, ending at D.
- Complete the rectangle ABCD.
- Draw perpendicular at meter divisions i.e 1, 2, 3, 4 and 5.
- Draw Horizontal line through the division along AD.
- Join D with end of the first division along AO (i.e 9).
- Through the remaining points i.e. 8, 7, 6 ... draw lines // to D9.

Diagonal Scale

$R.F = 3/200$